SOFTWARE ARCHITECTURE DOCUMENT

Online Exam Portal

Prepared with reference to the supplied project backlog

1. Introduction

1.1 Purpose

This Software Architecture Document (SAD) describes the proposed logical structure, main responsibilities, data flow, and quality considerations for the Online Exam Portal. It is based on the feature names visible in the supplied backlog screenshots. Where implementation details are not confirmed by those sources, this document describes the required responsibility rather than asserting a specific framework, database, or deployment design.

1.2 Scope

The portal supports user authentication and role-based access, examination creation and discovery, question-bank management, online examination attempts, countdown timing, answer saving, question navigation, scheduled access, automatic submission and evaluation, result viewing, student performance reports, examination statistics, secure data handling, network-loss consistency, backup/recovery, activity logs, and a responsive interface.

1.3 Intended Audience

This document is intended for project developers, testers, reviewers, and team members responsible for implementing or validating the listed backlog features.

2. Architectural Overview

The system can be understood as a set of cooperating logical layers. This is a logical view of responsibilities, not a claim that the repository uses a particular physical deployment or framework.

Logical Component	Responsibilities
User Interface / Presentation	Displays login, role-specific pages, available exams, exam questions, timer, results, reports and statistics. Must remain responsive and user-friendly.
Authentication and Authorization	Validates user credentials and restricts pages/actions according to the user's assigned role.
Examination Management	Creates and manages examinations, stores exam configuration and makes exams available according to configured access/schedule rules.
Question Bank Management	Adds, edits and deletes questions according to permissions; supplies questions for examinations.
Examination Attempt Management	Coordinates exam start, question navigation, answer selection/saving, countdown, submission and timeout handling.
Evaluation and Reporting	Evaluates objective questions using configured answer keys/rules and presents results, student performance reports and examination statistics.
Data Persistence and Recovery	Stores user, examination, question, attempt, answer and result data; supports consistency, backup and recovery according to implemented mechanisms.
Security and Activity Logging	Protects credentials and examination data, enforces authorization, and records supported activities for review.

3. Main Modules and Backlog Mapping

Backlog Item	Architectural Responsibility
Project Setup and Architecture	Application configuration and initialization.
US01 – User Login Authentication	Credential validation and authenticated session/user state.
US02 – Role-Based Access Control	Role-specific access to pages, data and actions.
US03 – View Available Exams	Retrieves and displays examinations available to a student.
US04 – Create Examination	Allows an authorized user to create and configure an examination.
US05 – Add Questions to Question Bank	Stores new questions and their required answer/options data.
US06 – Edit Questions in Question Bank	Updates existing question data.
US07 – Delete Questions from Question Bank	Removes questions subject to permissions and implemented rules.
US08 – Attend Examination Online	Starts an attempt and presents exam questions for answering.
US09 – Examination Countdown Timer	Tracks and displays remaining exam time.
US10 – Automatically Save Exam Answers	Persists selected answers according to the implemented save strategy.
US11 – Navigate Between Examination Questions	Moves between questions and preserves each question's answer state.
US12 – Restrict Exam Access Before Scheduled Time	Checks scheduled availability before allowing an attempt.
US13 – Automatically Submit Exam When Time Expires	Applies the configured end-of-time submission behavior.
US14 – Automatically Evaluate Objective Questions	Calculates objective-question scores using configured rules.

US15 – View Examination Results	Displays a student's result when result visibility permits.
US16 – Generate Student Performance Reports	Presents performance information based on available results.
US17 – Generate Examination Statistics	Summarizes examination/attempt data.
US18 – Secure User Credentials and Examination Data	Applies access restrictions and protects sensitive application data.
US19 – Maintain Data Consistency During Network Loss	Handles interruption and recovery without unintended answer/data inconsistency.
US20 – Backup and Recovery of Exam Data	Supports the project's defined backup and restoration process.
US21 – Maintain Activity Logs	Records supported actions for auditing/troubleshooting.
US22 – Responsive and User-Friendly Interface	Supports usable layouts across relevant viewport sizes.

4. Key Workflows

4.1 Login and Role-Based Access

1. A user opens the portal and submits credentials.
2. The authentication responsibility validates the credentials.
3. If authentication succeeds, the portal establishes the authenticated user state.
4. Authorization checks the user's role before protected pages, records or actions are made available.
5. Invalid credentials or unauthorized actions are rejected with appropriate feedback.

4.2 Examination Creation and Question Management

1. An authorized user opens examination management and enters the required examination details.
2. The examination management responsibility saves the configuration.
3. Questions are added to the question bank and can be edited or deleted by permitted users.
4. The examination is made available according to the configuration and scheduling rules implemented by the project.
5. Students see only examinations available to them.

4.3 Student Examination Attempt

1. The student selects an available examination.
2. The portal checks access permissions and any configured schedule.
3. The attempt opens with questions and a countdown timer.
4. The student answers questions and navigates between them; answers are saved according to the implemented strategy.
5. The student submits the attempt, or the portal applies the configured timeout behavior when time expires.
6. The attempt is evaluated according to supported marking rules, and results become visible according to release/access rules.

4.4 Reporting and Recovery

Submitted attempt and result data feed the result view, student performance reports and examination statistics. Activity logging records supported actions. Backup/recovery and network-interruption handling must preserve data according to the mechanisms implemented and configured for the project.

5. Data Responsibilities

Data Category	Information / Purpose
User data	Account identifier, credential-related data and role/authorization information.
Examination data	Exam details, schedule/availability configuration, duration and related settings.
Question-bank data	Question text, options/answer data and other fields supported by the implementation.
Attempt and answer data	Student attempt state, selected answers, timing/submission state and related records.
Evaluation and reporting data	Computed marks/results and the data used to generate performance reports and statistics.
Operational records	Activity logs and backup/recovery records where implemented.

6. Security and Quality Considerations

Quality Area	Design Consideration
Authentication and authorization	Validate credentials and check permissions for protected operations; do not rely only on hiding interface controls.
Data protection	Avoid exposing credentials or examination records to users without permission. Exact encryption and credential-storage mechanisms must be confirmed from the implementation.
Correctness of attempts	Keep question/answer associations consistent during navigation, saving, submission and evaluation.
Timing and submission	Use the configured exam duration and clearly defined timeout behavior; verify edge cases against the implemented workflow.

Resilience	Define and test expected behavior during temporary network loss and after recovery; do not assume unsaved client-side data is preserved unless implemented.
Recoverability	Document and test the actual backup/recovery process and its scope.
Observability	Record supported activities and ensure reports/statistics are derived from the appropriate stored data.
Usability	Keep exam actions clear and the interface usable at supported screen sizes.

7. Assumptions and Items to Confirm

The supplied backlog screenshots confirm the feature names, but do not identify the exact programming language, frontend/backend framework, database schema, API endpoints, authentication/session mechanism, hosting arrangement, or technical implementation of auto-save, backup, logging and network recovery. Those details should be filled in after inspecting the repository and its configuration. This document therefore describes logical responsibilities without inventing those implementation specifics.

The backlog screenshot shows “US05 – Add Questions to Question Bank” twice. This SAD treats it as one logical feature because no distinct requirement is visible for the duplicate entry.

8. Conclusion

The Online Exam Portal is organized around identity and access, examination/question management, student attempts, evaluation/reporting, and supporting security/reliability responsibilities. The architecture and workflows in this document map to the supplied backlog. Repository-specific technology and implementation decisions should be verified before this document is treated as a final as-built architecture.